UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Particular Integral, Type 1: Exponential Right-Hand Side

Complete study pack for class 003: the inverse differential operator, the important results for $\frac{1}{D}$ and $\frac{1}{D-a}$, the rule $\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}$ and its cases of failure, and the extension to constants, $\sinh ax$, $\cosh ax$ and a^x

Compiled from the PESU Unit 2 class 003 slide deck and AV-summary transcript, the Type 1 part of the class 004 problem transcript, the Unit 1 & 2 class-work sheet (problems 10–12) and eight past ESA papers. Generated 15 October 2026.

· How to use this pack

Topic 1 defines the inverse operator $\frac{1}{f(D)}$ and proves that $\frac{1}{f(D)}X$ is the particular integral. Topic 2 lists the important results ($\frac{1}{D}$ = integration, $\frac{1}{D-a}X = e^{ax}\int e^{-ax}X\,dx$). Topic 3 is the Type 1 rule and its failure ladder (Figure F3.1: learn it by heart). Topic 4 extends the rule to constants, $\sinh$, $\cosh$ and a^x and works the Type 1 part of every ESA question (table C4.2). The complementary function of each problem is written without comment: its rules are in the class 002 pack. The Type 2, 3 and 4 parts of the mixed ESA questions are solved in the class 004, 005 and 006 packs. Every topic has the same parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S003·8	Unit 2 deck 003 (“Particular integral of Type 1”), slide 8 (rendered .pptx)
T003 00:10:15	Unit 2 AV-summary transcript 003 (“Particular integral of standard functions – Type 1”, 21 min), segment start
T004a 00:02:06	Unit 2 AV-summary transcript in bucket 004 (“Particular integral Type 1 and Type 2”, 12.6 min); only its Type 1 part (00:00:00–00:06:04) and the constant term of its $\sin^2 x$ example are cited here. The class 004 pack cites the same file.
CW U2 P10	Unit 2 class-work problem 10
ESA Jan-24 s1 2d	Past paper: UE23MA141A Jan 2024 set 1, question 2.d

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2. 2 · Important results: $\frac{1}{D}$ and $\frac{1}{D-a}$
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1 The inverse differential operator and the particular integral

A non-homogeneous LDE $f(D)y = X$ ($X \neq 0$) has the general solution $y = \text{CF} + \text{PI}$. The CF solves $f(D)y = 0$ (class 002). The PI is found with the inverse differential operator $\frac{1}{f(D)}$: $\frac{1}{f(D)}X$ is the function of x, free from arbitrary constants, which gives back X when $f(D)$ acts on it. The theorem of slide 003·5 proves that $\frac{1}{f(D)}X$ is a particular solution, so the PI is $\frac{1}{f(D)}X$.

Key terms and definitions

Non-homogeneous LDE. $f(D)y = X$ where the right-hand side X is a non-zero function of the independent variable x. S003·6 · T003 00:03:22

Inverse differential operator. $\frac{1}{f(D)}(X)$ is that function of x which is free from arbitrary constants which when operated upon by $f(D)$ gives X. S003·4 · T003 00:01:38

Particular integral (PI). A particular solution of $f(D)y = X$ containing no arbitrary constant; it is given by $\frac{1}{f(D)}(X)$. Notation PI or y_p. S003·6 · S003·13 · T003 00:05:49

General (complete) solution. $y = \text{CF} + \text{PI}$, written $y = y_c + y_p$: the complementary function (general solution of $f(D)y = 0$, with n arbitrary constants) plus the particular integral. S003·6 · S003·12 · T003 00:00:43 · T003 00:05:32

> *"The inverse differential operator is denoted as $\frac{1}{f(D)}$. $\frac{1}{f(D)}(X)$ is that function of x which is free from arbitrary constants which when operated upon by $f(D)$ gives X. Thus, $f(D)$ and $\frac{1}{f(D)}$ are inverse operators. $f(D)\left\{\frac{1}{f(D)}X\right\} = X$."* S003·4

> *"Theorem: $\frac{1}{f(D)}(X)$ is the particular solution of $f(D)y = X$."* S003·5

> *"Consider the differential equation $f(D)y = X$. The general solution of the differential equation is y = Complementary function + Particular integral. The Particular Integral is given by $\frac{1}{f(D)}(X)$."* S003·6

Formulas

E1.1 · Inverse operators ESA ★★

$$f(D)\left\{\frac{1}{f(D)}X\right\} = X$$

gives	The defining property of $\frac{1}{f(D)}$.
applies	Any linear operator $f(D)$ with constant coefficients.
note	The order can be swapped: $\frac{1}{f(D)}\{f(D)X\} = X$ (T003 00:02:40), up to the CF.

S003·4 · T003 00:02:14 · T003 00:02:40

E1.2 · General solution of $f(D)y = X$ ESA ★★★

$$y = y_c + y_p = \text{CF} + \frac{1}{f(D)}X$$

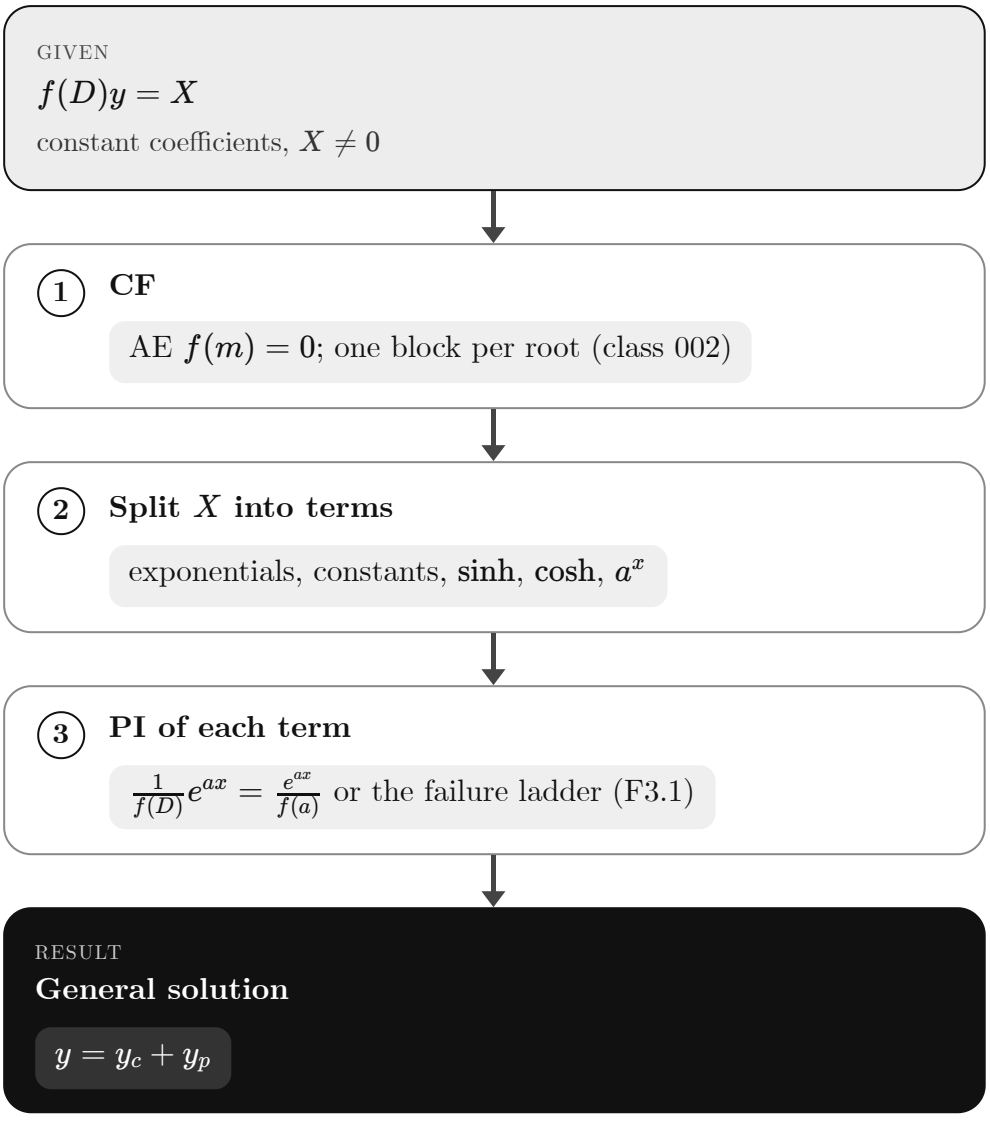
GIVES The complete solution.

APPLIES $X \neq 0$; the CF from the auxiliary equation (class 002).

NOTE CF has n arbitrary constants; the PI has none.

S003·6 · S003·12 · T003 00:05:32 · T003 00:06:07

Flowcharts



Remark. The CF roots tell you in advance which terms of X will fail: any e^{ax} whose a is an AE root.

Figure F1.1. Solving $f(D)y = X$ ESA ★★★

S003·6 · S003·11 · S003·12 · T003 00:17:57 · T003 00:20:36

Important points and why

1. Every non-homogeneous question needs two steps: the CF from the auxiliary equation, then the PI from $\frac{1}{f(D)}X$. Write $y = y_c + y_p$ at the end. ESA ★★★ S003·6 · T003 00:17:57 · T003 00:20:36
2. The PI contains no arbitrary constant; all n constants live in the CF. That is why $\frac{1}{D}$ is integration "without writing the constant". ESA ★★ S003·4 · T003 00:04:52 · T003 00:06:24
3. Proof of the theorem: put $y = \frac{1}{f(D)}X$ in $f(D)y = X$; by definition $f(D)\left[\frac{1}{f(D)}X\right] = X$, so it is a solution, and it has no arbitrary constants, so it is a particular solution. ESA ★★ S003·5 · T003 00:03:52 · T003 00:04:17
4. The lecturer calls the inverse operator "a very important concept" and the theorem "a very important statement". ESA ★★ T003 00:01:21 · T003 00:03:15

Why can the PI have no arbitrary constants?
The constants are already all in the CF (n of them for order n). Adding more would not give a more general solution, so $\frac{1}{f(D)}X$ is defined free of constants.
S003·4 · T003 00:04:52

SHORT-NOTE RECAP

For $f(D)y = X$ $(X \neq 0)$ the general solution is $y = \mathbf{CF} + \mathbf{PI}$. The inverse operator $\frac{1}{f(D)}X$ is the function of x, free from arbitrary constants, that gives X when $f(D)$ acts on it: $f(D)\{\frac{1}{f(D)}X\} = X$. Theorem: putting $y = \frac{1}{f(D)}X$ in the equation gives $X = X$, so $\frac{1}{f(D)}X$ is a particular solution, the PI.

2 Important results: $\frac{1}{D}$ and $\frac{1}{D-a}$

When $f(D)$ is simple the inverse operator can be written down: $\frac{1}{D}$ integrates (no constant), $\frac{1}{D^2}$ integrates twice, and $\frac{1}{D-a}X = e^{ax}\int e^{-ax}X\,dx$. A product of linear factors is handled one factor at a time or by partial fractions. These results prove the Type 1 rule and are the tool for right-hand sides with no shortcut (class 009, e^{e^x}).

Key terms and definitions

> *"Some important results:* $\frac{1}{D}(X) = \int X\,dx$ *where* X *is a function of* x*.* $\frac{1}{D-a}(X) = e^{ax}\int e^{-ax}X\,dx$ *where* X *is a function of* x*. If* $f(D) = (D-a)(D-b)$ *then* $\frac{1}{f(D)}(X) = \frac{1}{(D-a)(D-b)}(X) = \frac{1}{D-a}\left(\frac{1}{D-b}(X)\right)$*."* S003·7

Formulas

E2.1 · $\frac{1}{D}$ is integration ESA ★★

$$\frac{1}{D}X = \int X\,dx, \qquad \frac{1}{D^2}X = \iint X\,dx\,dx$$

GIVES $\frac{1}{f(D)}X$ for $f(D) = D$ or D^2.

APPLIES X any function of x.

NOTE No constant of integration: $\frac{1}{D}x^2 = \frac{x^3}{3}$.

S003·7 · T003 00:06:24 · T003 00:06:40 · T003 00:06:50

E2.2 · $\frac{1}{D-a}$ and $\frac{1}{D+a}$ ESA ★★

$$\frac{1}{D-a}X = e^{ax}\int e^{-ax}X\,dx, \qquad \frac{1}{D+a}X = e^{-ax}\int e^{ax}X\,dx$$

GIVES The PI for one linear factor and any X.

APPLIES a a constant.

NOTE "This is the formula you should remember" (T003 00:07:29). The $D+a$ form: replace a by $-a$ (lecture only). Used for e^{e^x} in class 009 (ESA Dec-23 3c).

S003·7 · T003 00:07:21 · T003 00:07:29 · T003 00:07:45 · ESA Dec-23 3c

E2.3 · Product of linear factors ESA ★

$$\frac{1}{(D-a)(D-b)}X = \frac{1}{D-a}\left(\frac{1}{D-b}X\right)$$

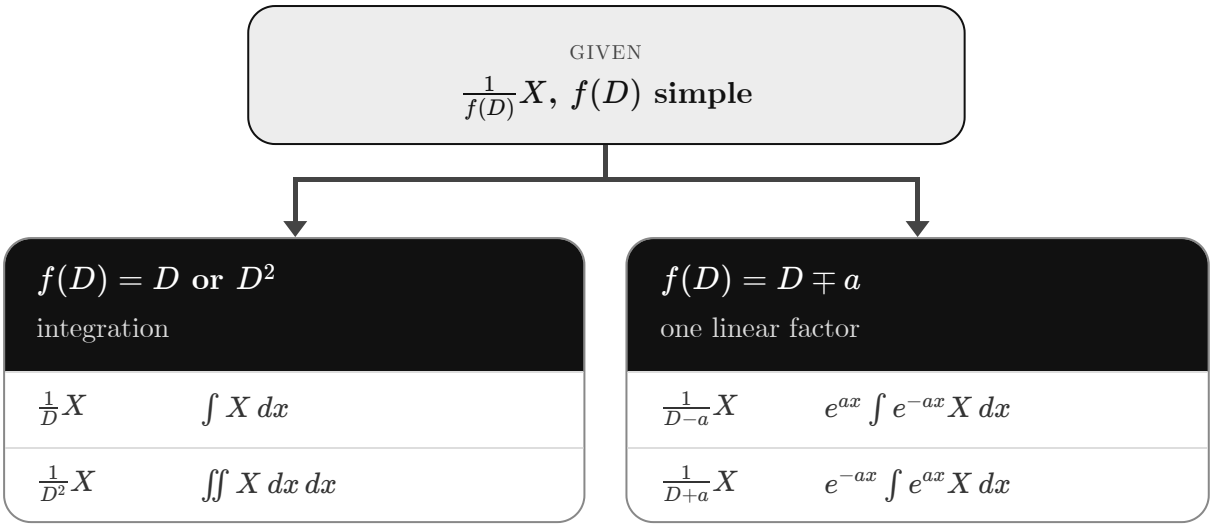
GIVES Apply the factors one at a time (innermost first).

APPLIES $f(D) = (D-a)(D-b)$.

NOTE Or by partial fractions: $\frac{1}{a-b}\left[\frac{1}{D-a} - \frac{1}{D-b}\right]X$ (T003 00:09:02).

S003·7 · T003 00:08:07 · T003 00:08:25 · T003 00:09:02

Flowcharts



Remark. No constants of integration: they all belong to the CF. For $(D-a)(D-b)$ apply $\frac{1}{D-b}$ first and then $\frac{1}{D-a}$, or split into partial fractions and use E2.2 on each.

Figure F2.1. Important results for simple $f(D)$ ESA ★★

S003·7 · T003 00:06:24 · T003 00:07:29 · T003 00:09:02

Important points and why

1. The operator always precedes the function it acts on: write $\frac{1}{f(D)}X$, never $X \cdot \frac{1}{f(D)}$. ESA ★★ T003 00:09:20
2. For $(D-a)(D-b)$ apply $\frac{1}{D-b}$ first, then $\frac{1}{D-a}$ to the result; or split into partial fractions and use E2.2 on each. ESA ★ S003·7 · T003 00:08:25 · T003 00:09:02
3. E2.2 explains the failure case: $\frac{1}{D-a}e^{ax} = e^{ax}\int e^{-ax}e^{ax}dx = xe^{ax}$, the factor x of E3.2. ESA ★★ ADDED FOR CLARITY S003·7

Exam trap. $\frac{1}{D^2}X$ means integrate twice. The lecture's "2 times integration" (T003 00:07:03) is not $2\int X\,dx$. S003·7 · T003 00:07:03

Worked examples

W2.1 · $\frac{1}{D}$ and $\frac{1}{D^2}$ of x^2 ESA ★

GIVEN Find $\frac{1}{D}x^2$ and $\frac{1}{D^2}x^2$.

STEP 1 $\frac{1}{D}x^2 = \int x^2 dx = \frac{x^3}{3}$ (no constant).

STEP 2 $\frac{1}{D^2}x^2 = \frac{1}{D}\frac{x^3}{3} = \frac{x^4}{12}$.

ANSWER $\boxed{\frac{x^3}{3} \text{ and } \frac{x^4}{12}}$

CHECK $D\frac{x^3}{3} = x^2$, $D^2\frac{x^4}{12} = x^2$ ✓.

T003 00:06:40 · T003 00:06:50

W2.2 · $\frac{1}{D-a}$ gives the failure case ESA ★★

GIVEN Find $\frac{1}{D-1}e^x$ and $\frac{1}{D-1}e^{3x}$ with E2.2.

STEP 1 $\frac{1}{D-1}e^x = e^x \int e^{-x}e^x dx = e^x \int 1\,dx = xe^x$.

STEP 2 $\frac{1}{D-1}e^{3x} = e^x \int e^{2x} dx = e^x \cdot \frac{e^{2x}}{2} = \frac{e^{3x}}{2}$.

STEP 3 Type 1 agrees: $f(3) = 2$ gives $\frac{e^{3x}}{2}$; $f(1) = 0$, $f'(1) = 1$ gives $\frac{xe^x}{1}$.

ANSWER $\boxed{xe^x \text{ and } \frac{1}{2}e^{3x}}$

CHECK $(D-1)xe^x = e^x + xe^x - xe^x = e^x$ ✓; sympy agrees.

ADDED FOR CLARITY S003·7 · T003 00:07:29

W2.3 · Two factors by partial fractions ESA ★

GIVEN Find $\frac{1}{(D-1)(D-2)}e^{3x}$ by partial fractions and E2.2.

STEP 1 $\frac{1}{(D-1)(D-2)} = \frac{1}{D-2} - \frac{1}{D-1}$.

STEP 2 $\frac{1}{D-2}e^{3x} = e^{2x}\int e^x dx = e^{3x}$; $\frac{1}{D-1}e^{3x} = \frac{e^{3x}}{2}$ (W2.2).

STEP 3 Difference: $e^{3x} - \frac{e^{3x}}{2}$.

ANSWER $\boxed{\frac{1}{2}e^{3x}}$

CHECK Type 1: $f(3) = (2)(1) = 2$, so $\frac{e^{3x}}{2}$ ✓.

ADDED FOR CLARITY S003·7 · T003 00:09:02

SHORT-NOTE RECAP

$\frac{1}{D}X = \int X\,dx$ and $\frac{1}{D^2}X = \iint X\,dx\,dx$ (no constants). $\frac{1}{D-a}X = e^{ax}\int e^{-ax}X\,dx$, $\frac{1}{D+a}X = e^{-ax}\int e^{ax}X\,dx$. For $(D-a)(D-b)$ apply the factors one at a time or use partial fractions. Write the operator before the function.

3 Type 1: $X = e^{ax}$ and the cases of failure

$\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}$ provided $f(a) \neq 0$: replace D by a. When $f(a) = 0$ (a is a root of the auxiliary equation) multiply by x and differentiate the denominator: $\frac{xe^{ax}}{f'(a)}$; if also $f'(a) = 0$, $\frac{x^2e^{ax}}{f''(a)}$, and so on. Every ESA question with an exponential term uses this ladder; three of the eight papers ask the double failure $\frac{8e^{2x}}{(D-2)^2}$.

Key terms and definitions

Case of failure. $f(a) = 0$: the rule $\frac{e^{ax}}{f(a)}$ divides by zero. It happens exactly when $D - a$ is a factor of $f(D)$, i.e. a is a root of the auxiliary equation. S003·8 · T003 00:11:22

> *"Rules to find particular integral. Type (i): $X = e^{ax}$. $\frac{1}{f(D)}(e^{ax}) = \frac{e^{ax}}{f(a)}$ provided $f(a) \neq 0$. Cases of failure: when $f(a) = 0$, then $\frac{1}{f(D)}(e^{ax}) = x \cdot \frac{e^{ax}}{f'(a)}$ provided $f'(a) \neq 0$. When $f'(a) = 0$, then $\frac{1}{f(D)}(e^{ax}) = x^2 \cdot \frac{e^{ax}}{f''(a)}$ provided $f''(a) \neq 0$."* S003·8

> *"If $f(a) = 0$, then P.I $= x\frac{e^{ax}}{f'(a)}$, provided $f'(a) \neq 0$. If $f'(a) = 0$, then P.I $= x^2\frac{e^{ax}}{f''(a)}$, provided $f''(a) \neq 0$, similarly, it proceeds on when $f''(a) = 0$....."* S003·13

Formulas

E3.1 · Type 1 rule ESA ★★★

$$\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}, \qquad f(a) \neq 0$$

GIVES The PI of an exponential.

APPLIES $X = e^{ax}$, a positive, negative or zero.

NOTE Replace D by a. For e^{-ax} replace D by $-a$: $\frac{e^{-ax}}{f(-a)}$ (T003 00:10:51).

S003·8 · S003·13 · T003 00:10:15 · T003 00:10:51 · T004a 00:00:41

E3.2 · First case of failure ESA ★★★

$$f(a) = 0: \quad \frac{1}{f(D)}e^{ax} = x\,\frac{e^{ax}}{f'(a)}, \qquad f'(a) \neq 0$$

GIVES The PI when a is a simple root of $f(m) = 0$.

APPLIES $D - a$ is a factor of $f(D)$ once.

NOTE $f'(D) = \frac{d}{dD}f(D)$; x is written outside the operator (T003 00:11:51).

S003·8 · S003·13 · T003 00:11:35 · T003 00:12:08 · T004a 00:01:20 · ESA Jan-24 s2 2d

E3.3 · Second case of failure ESA ★★★

$$f(a)=f'(a)=0: \quad \frac{1}{f(D)}e^{ax}=x^2\,\frac{e^{ax}}{f''(a)}, \qquad f''(a)\neq 0$$

GIVES The PI when a is a double root.

APPLIES $(D-a)^2$ divides $f(D)$.

NOTE ESA Dec-23 3a: $\frac{8e^{2x}}{(D-2)^2}=\frac{8x^2e^{2x}}{2}=4x^2e^{2x}$.

S003·8 · S003·13 · T003 00:12:53 · T004a 00:01:41 · ESA Dec-23 3a

E3.4 · Third case of failure ESA ★★

$$f(a)=f'(a)=f''(a)=0: \quad \frac{1}{f(D)}e^{ax}=x^3\,\frac{e^{ax}}{f'''(a)}$$

GIVES The PI when a is a triple root.

APPLIES $(D-a)^3$ divides $f(D)$; "and so on".

NOTE CW P12: $\frac{4e^{-x}}{(D+1)^3(D-2)}=\frac{4x^3e^{-x}}{f'''(-1)}=-\frac{2}{9}x^3e^{-x}$.

TRANSCRIPT ONLY T003 00:13:19 · S003·13 · CW U2 P12

E3.5 · Root of multiplicity k (shortcut) ESA ★★

$$f(D)=(D-a)^k\varphi(D),\ \varphi(a)\neq 0: \quad \frac{1}{f(D)}e^{ax}=\frac{x^k\,e^{ax}}{k!\,\varphi(a)}$$

GIVES The same answer as the ladder, in one step.

APPLIES The factorised $f(D)$ is known.

NOTE $f^{(k)}(a)=k!\,\varphi(a)$. Jan-24 s2 2d: $\varphi(D)=(D-1)(D-3)$, $\varphi(2)=-1$, PI $=-xe^{2x}$.

ADDED FOR CLARITY ESA Jan-24 s2 2d · ESA Dec-23 3a

Graphs and diagrams

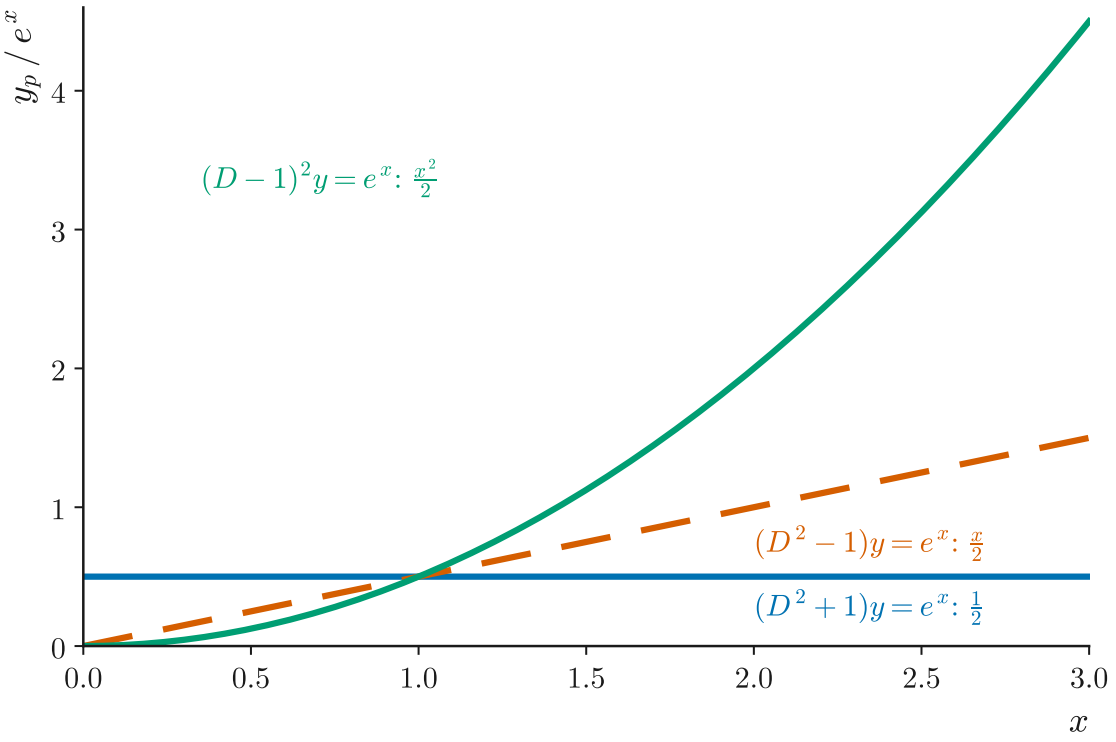


Figure G3.1. The failure ladder: the PI divided by e^x. ESA ★★★

ADDED FOR CLARITY S003·8 · S003·13 · T003 00:12:39

How to read it. Three equations with the same forcing e^x ($a = 1$). If 1 is not a root ($D^2 + 1$, $f(1) = 2$) the PI is a constant multiple $\frac{1}{2}e^x$ (blue, flat ratio). If 1 is a simple root ($D^2 - 1$, $f'(1) = 2$) the ratio grows like $\frac{x}{2}$ (vermilion dashed). If 1 is a double root ($(D-1)^2$, $f''(1) = 2$) it grows like $\frac{x^2}{2}$ (green). Each failure adds one power of x, a resonance-like growth.

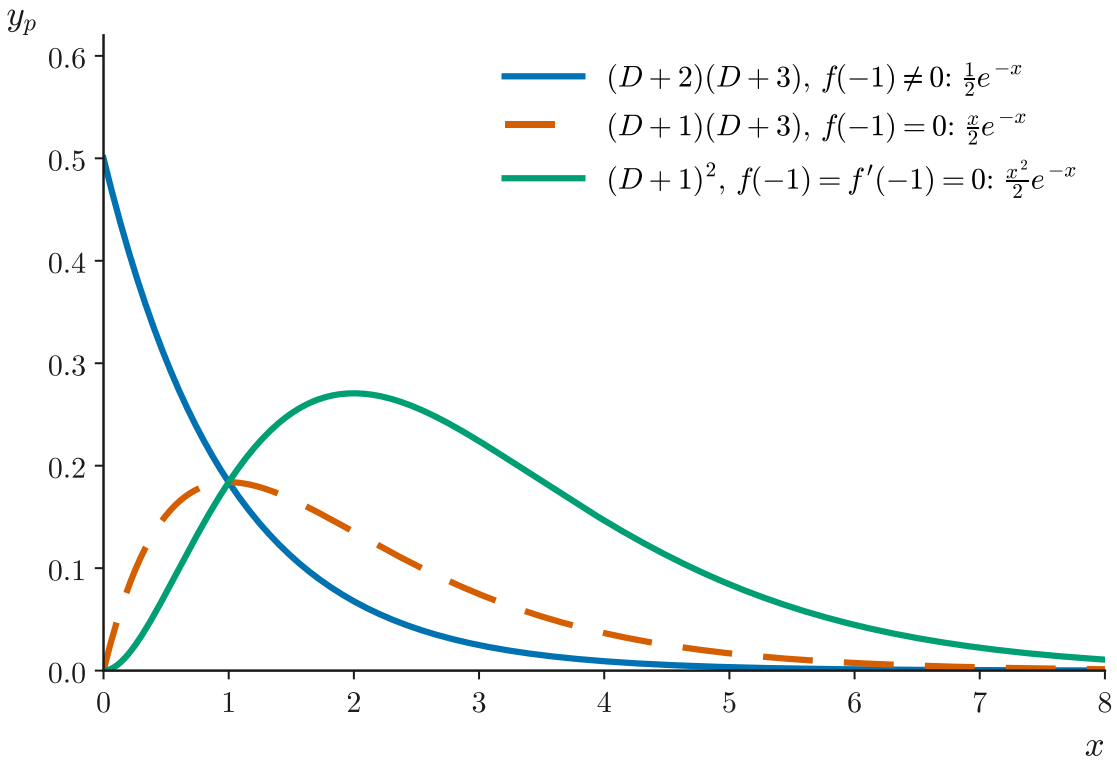
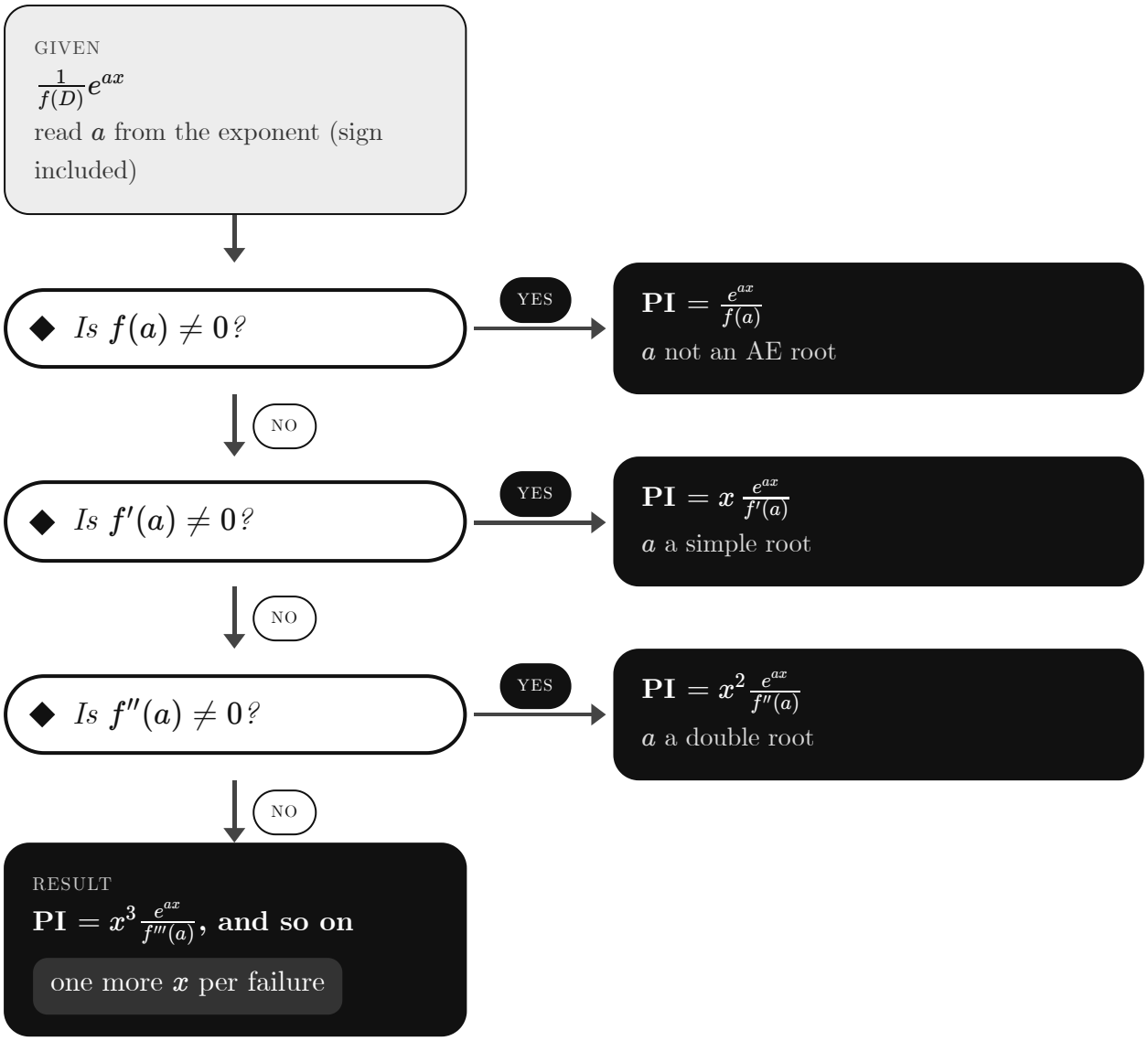


Figure G3.2. Forcing e^{-x}: e^{-x}, xe^{-x} and x^2e^{-x} particular integrals. ESA ★★

ADDED FOR CLARITY S003·8 · S003·15

How to read it. For $(D+2)(D+3)$, $(D+1)(D+3)$ and $(D+1)^2$ with $X=e^{-x}$ the PIs are $\frac{1}{2}e^{-x}$ (blue), $\frac{x}{2}e^{-x}$ (vermilion dashed) and $\frac{x^2}{2}e^{-x}$ (green). Without failure the PI just copies the shape of X; with a failure it rises before it decays, because e^{-x} is already in the CF. Compare slide 003·15, $\frac{x^2}{2}e^{-2x}$.

Flowcharts



Remark. Derivatives are taken with respect to D, then $D = a$. Shortcut (added for clarity): $f(D) = (D-a)^k\varphi(D)$ gives $\frac{x^k e^{ax}}{k!\,\varphi(a)}$.

Exam trap. Keep the constant factor: $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$.

Figure F3.1. The Type 1 ladder for $\frac{1}{f(D)}e^{ax}$ ESA ★★★

S003·8 · S003·13 · T003 00:11:35 · T003 00:12:53 · T003 00:13:19 · T004a 00:01:01

Type 1 and its cases of failure

Condition	a as AE root	PI of e^{ax}	Example
$f(a) \neq 0$	not a root	$\frac{e^{ax}}{f(a)}$	$\frac{e^{x}}{D^2-3D-4} = -\frac{e^{x}}{6}$
$f(a) = 0,\ f'(a) \neq 0$	simple	$x\frac{e^{ax}}{f'(a)}$	$\frac{4e^{4x}}{D^2-3D-4} = \frac{4}{5}xe^{4x}$
$f'(a) = 0,\ f''(a) \neq 0$	double	$x^2\frac{e^{ax}}{f''(a)}$	$\frac{e^{-2x}}{(D+2)^2} = \frac{x^2}{2}e^{-2x}$
$f''(a) = 0,\ f'''(a) \neq 0$	triple	$x^3\frac{e^{ax}}{f'''(a)}$	$\frac{4e^{-x}}{(D+1)^3(D-2)} = -\frac{2}{9}x^3e^{-x}$

S003·8 · S003·13 · T003 00:13:19 · S003·15 · CW U2 P10 · CW U2 P12

Important points and why

1. To apply Type 1, read a from the exponent: e^{-2x} has $a = -2$, a constant has $a = 0$, e^{x} has $a = 1$. Then replace every D in $f(D)$ by a. ESA ★★★ S003·8 · S003·15 · T004a 00:00:41 · T004a 00:02:59
2. $f(a) = 0$ exactly when $D - a$ is a factor of $f(D)$, i.e. when e^{ax} is already in the CF. Compare the exponent with the AE roots before substituting. ESA ★★★ T003 00:11:22 · T003 00:12:39
3. At each failure, multiply by one more x (outside the operator) and differentiate $f(D)$ once more with respect to D; stop at the first non-zero derivative. ESA ★★★ S003·8 · S003·13 · T003 00:12:08 · T004a 00:03:44
4. The power of x equals the multiplicity of a as a root: simple root x, double root x^2, triple root x^3 (Figure G3.1). ESA ★★★ ADDED FOR CLARITY T003 00:12:39 · T003 00:13:19
5. ESA: the double failure $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$ appears in three papers (Dec-23, May-24, Jun-24); the single failure $\frac{e^{2x}}{(D-1)(D-2)(D-3)} = -xe^{2x}$ in Jan-24 set 2. ESA ★★★ ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a · ESA Jan-24 s2 2d

Why does replacing D by a work?
$D^k e^{ax} = a^k e^{ax}$, so $f(D)e^{ax} = f(a)e^{ax}$. Applying $f(D)$ to $\frac{e^{ax}}{f(a)}$ therefore gives e^{ax}, which is exactly the definition of $\frac{1}{f(D)}e^{ax}$.
ADDED FOR CLARITY S003·4 · S003·8

Why an extra x when $f(a) = 0$?
Then e^{ax} solves $f(D)y = 0$ (it is in the CF), so no multiple of e^{ax} can give a non-zero right-hand side. xe^{ax} is the next independent function: $\frac{1}{D-a}e^{ax} = e^{ax}\int 1\,dx = xe^{ax}$ (E2.2).
ADDED FOR CLARITY S003·7 · T003 00:11:22

Exam trap. Check $f(a)$ before writing $\frac{e^{ax}}{f(a)}$: $\frac{e^{2x}}{(D-2)^2}$ is not $\frac{e^{2x}}{0}$. Compare a with the AE roots you just found for the CF. ADDED FOR CLARITY S003·8 · ESA Dec-23 3a

Exam trap. $f'(a)$ means differentiate $f(D)$ with respect to D, then put $D = a$; do not differentiate e^{ax}. The x goes outside: $x\frac{e^{ax}}{f'(a)}$. T003 00:11:51 · T003 00:12:08

Exam trap. For e^{-2x}, $a=-2$ (keep the sign): $f(-2)$ for D^2+4D+4 is $4-8+4=0$, a failure.

S003·15 · T004a 00:03:25

Exam trap. A constant factor stays: $\frac{8e^{2x}}{(D-2)^2}=8\cdot\frac{x^2e^{2x}}{2}=4x^2e^{2x}$, not $\frac{x^2e^{2x}}{2}$. ADDED FOR CLARITY ESA Dec-23 3a

Worked examples

W3.1 · Double failure: $y''+4y'+4y=e^{-2x}$ (slide 003·15) ESA ★★★

GIVEN Find the PI of $y''+4y'+4y=e^{-2x}$.

STEP 1 $a=-2$, $f(D)=D^2+4D+4$: $f(-2)=4-8+4=0$, failure.

STEP 2 $f'(D)=2D+4$: $f'(-2)=0$, failure again.

STEP 3 $f''(D)=2$: $f''(-2)=2\neq 0$, so PI $=\frac{x^2e^{-2x}}{f''(-2)}$.

ANSWER $\boxed{\text{PI}=\frac{x^2e^{-2x}}{2};\ y=(c_1+c_2x)e^{-2x}+\frac{x^2}{2}e^{-2x}}$

CHECK sympy dsolve gives $(C_1+x(C_2+\frac{x}{2}))e^{-2x}$ ✓. -2 is a double root of $(m+2)^2$, hence x^2.

S003·15 · T004a 00:02:59 · T004a 00:04:02 · T004a 00:04:22

W3.2 · One term fails: $y''-3y'-4y=e^x+4e^{4x}$ (CW P10) ESA ★★★

GIVEN Solve $\frac{d^2y}{dx^2}-3\frac{dy}{dx}-4y=e^x+4e^{4x}$.

STEP 1 AE $m^2-3m-4=(m-4)(m+1)=0$: $y_c=c_1e^{-x}+c_2e^{4x}$.

STEP 2 $\frac{e^x}{D^2-3D-4}$: $f(1)=1-3-4=-6$, so $-\frac{e^x}{6}$.

STEP 3 $\frac{4e^{4x}}{D^2-3D-4}$: $f(4)=16-12-4=0$; $f'(D)=2D-3$, $f'(4)=5$, so $\frac{4xe^{4x}}{5}$.

ANSWER $\boxed{y=c_1e^{-x}+c_2e^{4x}-\frac{e^x}{6}+\frac{4}{5}xe^{4x}}$

CHECK sympy dsolve and checkodesol ✓. The CW sheet misprints the answer as $(c_1e^{-x}+c_2e^{4x}x)-\frac{e^x}{6}+\frac{4xe^x}{5}$; substituting $\frac{4xe^x}{5}$ fails.

CW U2 P10

W3.3 · Triple root: $(D-1)^3 y = e^x$ ESA ★

GIVEN Find $\frac{1}{(D-1)^3} e^x$.

STEP 1 $f(1) = 0$; $f'(D) = 3(D-1)^2$, $f'(1) = 0$; $f''(D) = 6(D-1)$, $f''(1) = 0$.

STEP 2 $f'''(D) = 6$: PI $= \frac{x^3 e^x}{f'''(1)} = \frac{x^3 e^x}{6}$.

STEP 3 Shortcut E3.5: $k = 3$, $\varphi = 1$: $\frac{x^3 e^x}{3!}$.

ANSWER $\boxed{\frac{x^3 e^x}{6}}$

CHECK Substituting back into $(D-1)^3 y$ gives e^x (sympy) ✓.

ADDED FOR CLARITY T003 00:13:19

W3.4 · Single failure, e^{2x} part of ESA Jan-24 s2 2d ESA ★★★

GIVEN Find the PI of e^{2x} in $(D^3 - 6D^2 + 11D - 6)y = e^{2x} + x$.

STEP 1 $f(2) = 8 - 24 + 22 - 6 = 0$: failure (2 is a root; $f(D) = (D-1)(D-2)(D-3)$).

STEP 2 $f'(D) = 3D^2 - 12D + 11$: $f'(2) = 12 - 24 + 11 = -1 \neq 0$.

STEP 3 $\mathrm{PI}_1 = \frac{x e^{2x}}{-1}$. (Shortcut: $\varphi(2) = (2-1)(2-3) = -1$.)

ANSWER $\boxed{\mathrm{PI}_1 = -x e^{2x}}$

CHECK $f(D)(-x e^{2x}) = e^{2x}$ (sympy) ✓. The x term (Type 3) and the full solution are in the class 005 pack.

ESA Jan-24 s2 2d

W3.5 · Double failure, $8e^{2x}$ part of ESA Dec-23 3a ESA ★★★

GIVEN Find the PI of $8e^{2x}$ in $(D-2)^2 y = 8(e^{2x} + \sin 2x)$.

STEP 1 $f(D) = (D-2)^2$: $f(2) = 0$.

STEP 2 $f'(D) = 2(D-2)$: $f'(2) = 0$.

STEP 3 $f''(D) = 2$: $\mathrm{PI}_1 = 8 \cdot \frac{x^2 e^{2x}}{2}$.

ANSWER $\boxed{\mathrm{PI}_1 = 4x^2 e^{2x}}$

CHECK $(D-2)^2(4x^2 e^{2x}) = 8e^{2x}$ (sympy) ✓. With the Type 2 part, $\frac{8\sin 2x}{(D-2)^2} = \cos 2x$, the full answer is $y = (c_1 + c_2 x)e^{2x} + 4x^2 e^{2x} + \cos 2x$ (class 004 pack).

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a

W3.6 · No failure, e^{4x} part of ESA Jul-23 3b ESA ★★★

GIVEN Find the PI of e^{4x} in $(D^2-5D+6)y = e^{4x}+\sin x + x^2$.

STEP 1 AE roots $2, 3$; $a = 4$ is not a root.

STEP 2 $f(4) = 16 - 20 + 6 = 2$.

ANSWER $\boxed{\text{PI}_1 = \frac{e^{4x}}{2}}$

CHECK $(D^2-5D+6)\frac{e^{4x}}{2} = \frac{16-20+6}{2}e^{4x} = e^{4x}$ ✓. The other parts are in the class 005 pack.

ESA Jul-23 3b

Past-paper questions

ESA Dec 2023 (7 marks) UE22MA141A Q3.a (= May 2024 Q3.b, June 2024 Q2.a) · solved in W3.5

Solve $(D-2)^2y = 8(e^{2x}+\sin 2x)$. Type 1 part: $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$ (double failure). Full solution: class 004 pack.

ESA Jul 2023 (7 marks) UE22MA141A Q3.b · solved in W3.6

Solve $(D^2-5D+6)y = e^{4x}+\sin x + x^2$. Type 1 part: $\frac{e^{4x}}{2}$. Full solution: class 005 pack.

ESA Jan 2024 (5 marks) UE23MA141A set 2, Q2.d · solved in W3.4

Solve $(D^3-6D^2+11D-6)y = e^{2x}+x$. Type 1 part: $-xe^{2x}$ (single failure). Full solution: class 005 pack.

SHORT-NOTE RECAP

$\frac{1}{f(D)}e^{ax} = \frac{e^{ax}}{f(a)}$ if $f(a)\neq 0$. If $f(a)=0$: $\frac{xe^{ax}}{f'(a)}$; if $f'(a)=0$ too: $\frac{x^2e^{ax}}{f''(a)}$; then $\frac{x^3e^{ax}}{f'''(a)}$, and so on. The power of x is the multiplicity of a as an AE root. Examples: $\frac{e^{-2x}}{D^2+4D+4} = \frac{x^2e^{-2x}}{2}$; $\frac{8e^{2x}}{(D-2)^2} = 4x^2e^{2x}$ (ESA ×3).

4 Extensions: constants, $\sinh ax$, $\cosh ax$ and a^x

Any right-hand side that is a sum of exponentials uses the same rule term by term. A constant is $k = ke^{0x}$, so $\frac{k}{f(D)} = \frac{k}{f(0)}$. $\sinh ax$ and $\cosh ax$ split into $\frac{1}{2}(e^{ax} \mp e^{-ax})$. $a^x = e^{x \log a}$, so D is replaced by $\log a$. The ESA asks 2^{-x} twice and the constant from $\sin^2$ twice; the full mixed questions are in the class 004–006 packs.

Key terms and definitions

> *"Extension of Type (i). When $X = K$: $\frac{K}{f(D)} = K\frac{e^{0x}}{f(D)} = \frac{K}{f(0)}$. When $X = \sinh ax$: $\frac{\sinh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}\right]$. When $X = \cosh ax$: $\frac{\cosh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} + \frac{e^{-ax}}{f(-a)}\right]$. When $X = a^x$: $\frac{a^x}{f(D)} = \frac{e^{(\log a)x}}{f(D)} = \frac{e^{(\log a)x}}{f(\log a)}$. Thus, $\frac{a^x}{f(D)} = \frac{a^x}{f(\log a)}$."* S003·9 · S003·10

Formulas

E4.1 · Constant right-hand side — ESA ★★

$$\frac{k}{f(D)} = k\,\frac{e^{0x}}{f(D)} = \frac{k}{f(0)}, \qquad f(0) \neq 0$$

GIVES The PI of a constant.

APPLIES $X = k$.

NOTE If $f(0) = 0$: $\frac{xk}{f'(0)}$, then $\frac{x^2 k}{f''(0)}$, ... (T003 00:14:25). Slide 003·14: $\frac{1}{D^2+2D+5} = \frac{1}{5}$.

S003·9 · S003·14 · T003 00:13:50 · T003 00:14:25 · T004a 00:02:06 · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

E4.2 · $\sinh ax$ — ESA ★★

$$\frac{\sinh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}\right]$$

GIVES The PI of $\sinh ax$.

APPLIES $f(a) \neq 0$, $f(-a) \neq 0$; otherwise use the failure ladder on that term.

NOTE $\sinh ax = \frac{e^{ax} - e^{-ax}}{2}$. Slide 003·16: $2\sinh x = e^x - e^{-x}$.

S003·9 · T003 00:14:55 · T003 00:15:19 · CW U2 P11

E4.3 · $\cosh ax$ — ESA ★★

$$\frac{\cosh ax}{f(D)} = \frac{1}{2}\left[\frac{e^{ax}}{f(a)} + \frac{e^{-ax}}{f(-a)}\right]$$

GIVES The PI of $\cosh ax$.

APPLIES $f(\pm a) \neq 0$; otherwise use the ladder.

NOTE $\cosh ax = \frac{e^{ax} + e^{-ax}}{2}$. CW P12: $e^{-x}\cosh x = \frac{1+e^{-2x}}{2}$.

S003·10 · T003 00:15:37 · CW U2 P12

E4.4 · a^x and a^{-x} ESA ★★

$$\frac{a^x}{f(D)} = \frac{a^x}{f(\log a)}, \qquad \frac{a^{-x}}{f(D)} = \frac{a^{-x}}{f(-\log a)}$$

GIVES The PI of a power of a constant.

APPLIES $a > 0$, $f(\pm \log a) \neq 0$.

NOTE $a^x = e^{x \log a}$: the exponent is $\log a$ (natural log). ESA Jan-24 s1 2d and s4 2c ask 2^{-x}.

S003·10 · T003 00:15:54 · T003 00:16:34 · T003 00:16:48 · ESA Jan-24 s1 2d · ESA Jan-24 s4 2c

Graphs and diagrams

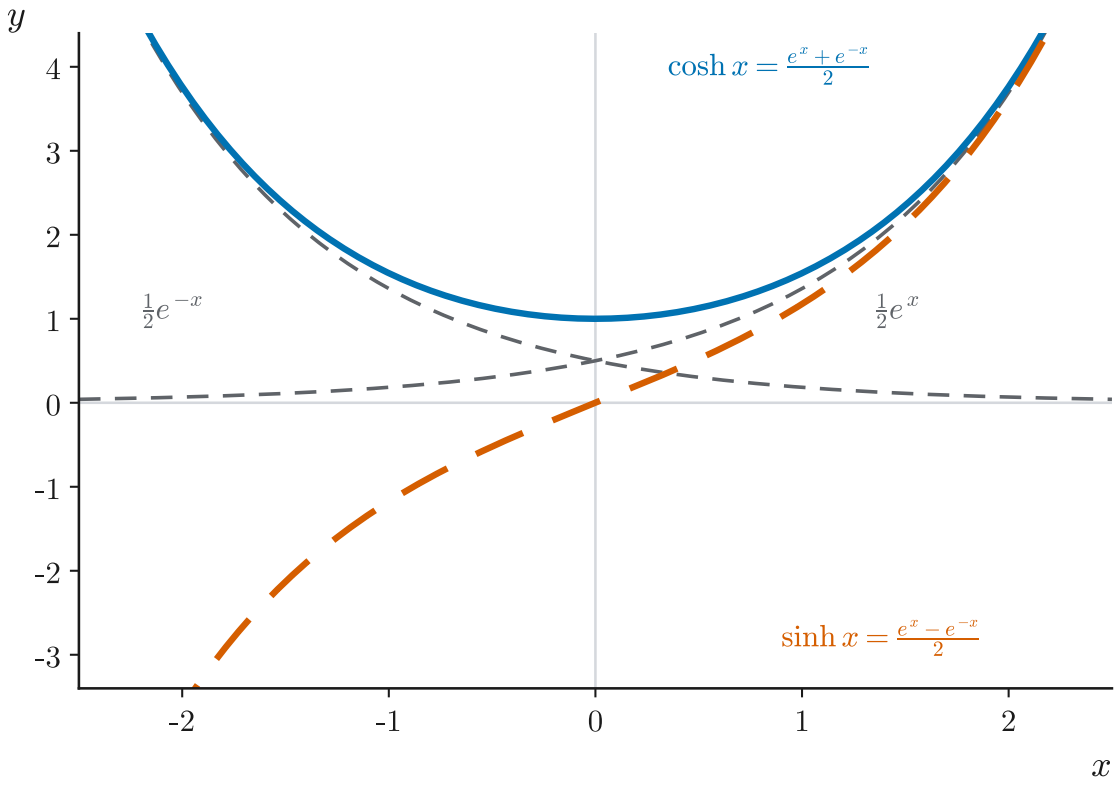


Figure G4.1. $\sinh x$ and $\cosh x$ are sums of two exponentials. ESA ★★

ADDED FOR CLARITY S003·9 · S003·10

How to read it. $\cosh x$ (blue) is the sum and $\sinh x$ (vermilion dashed) the difference of $\frac{1}{2}e^x$ and $\frac{1}{2}e^{-x}$ (grey). So $\frac{1}{f(D)}$ acts on each half separately, with $D \to a$ and $D \to -a$ (E4.2, E4.3).

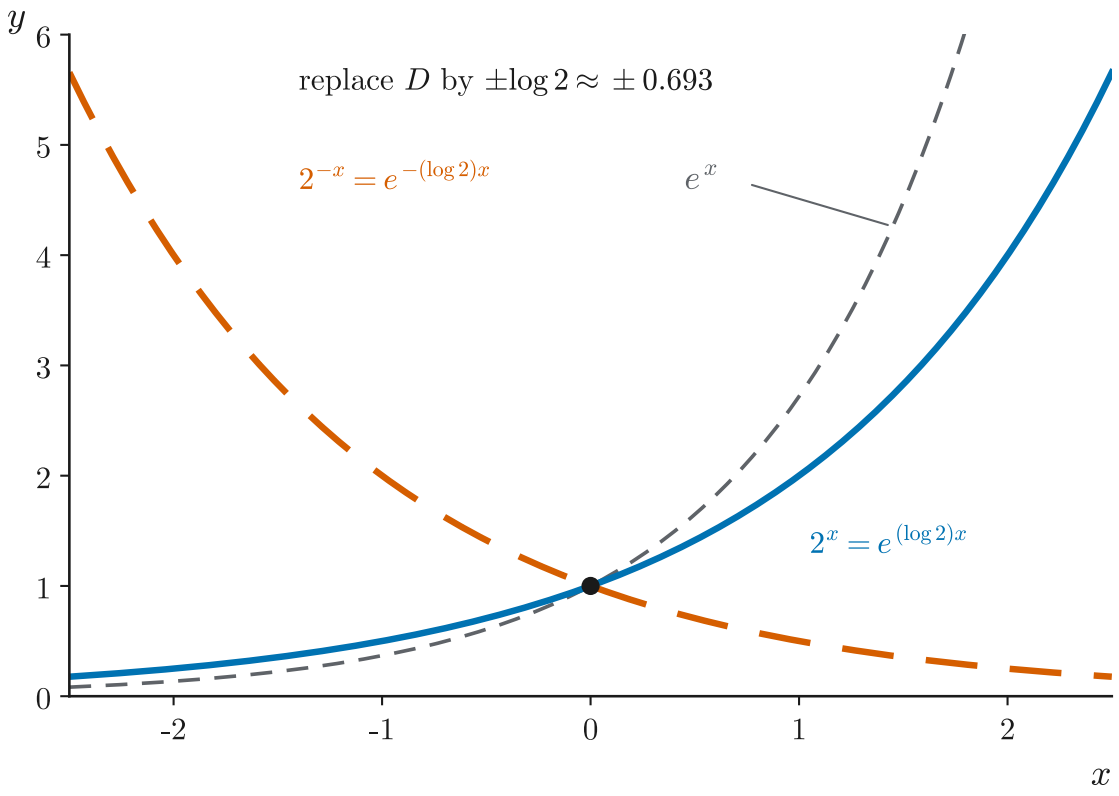
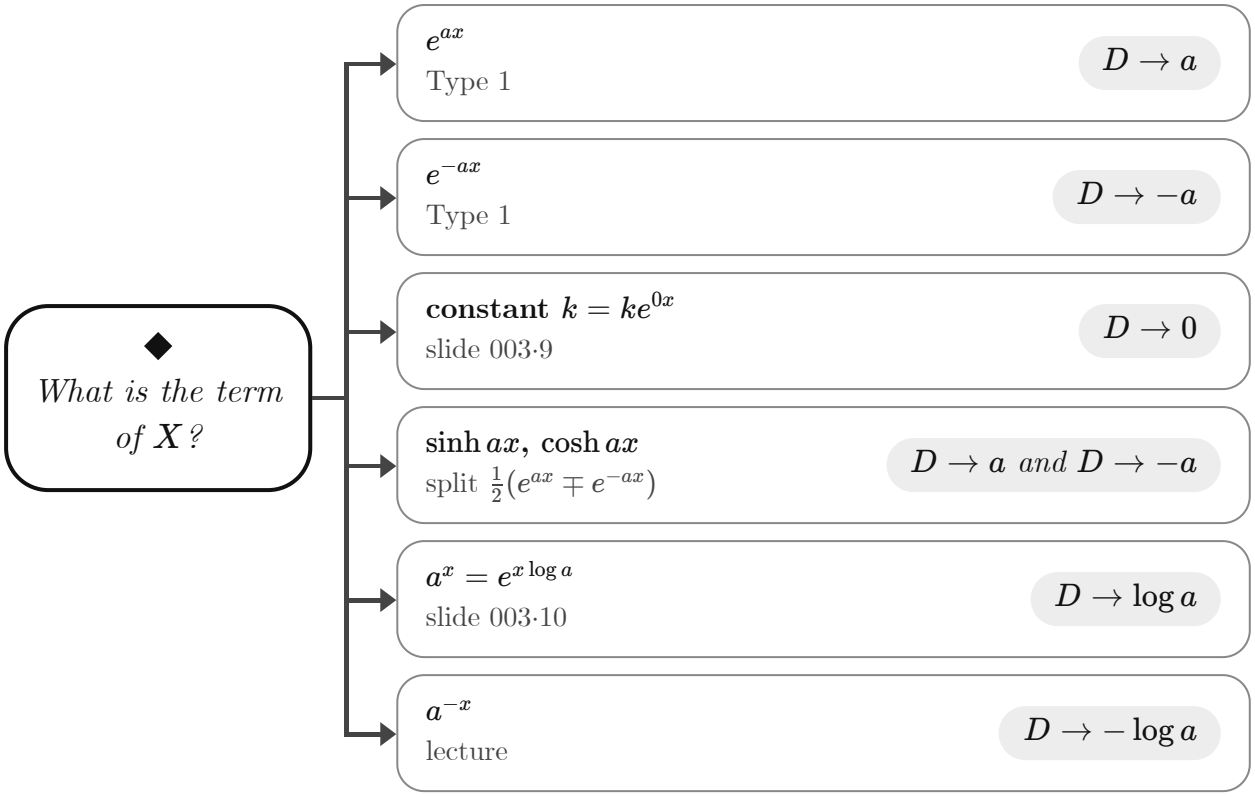


Figure G4.2. $a^x = e^{x \log a}$: 2^x and 2^{-x}. ESA ★★

ADDED FOR CLARITY S003·10 · T003 00:16:11 · ESA Jan-24 s1 2d

How to read it. 2^x (blue) is an exponential with exponent $\log 2 \approx 0.693$, flatter than e^x (grey). 2^{-x} (vermilion dashed) has exponent $-\log 2$. All pass through $(0, 1)$. In the Type 1 rule, D is replaced by the exponent: $\pm \log 2$, never by 2.

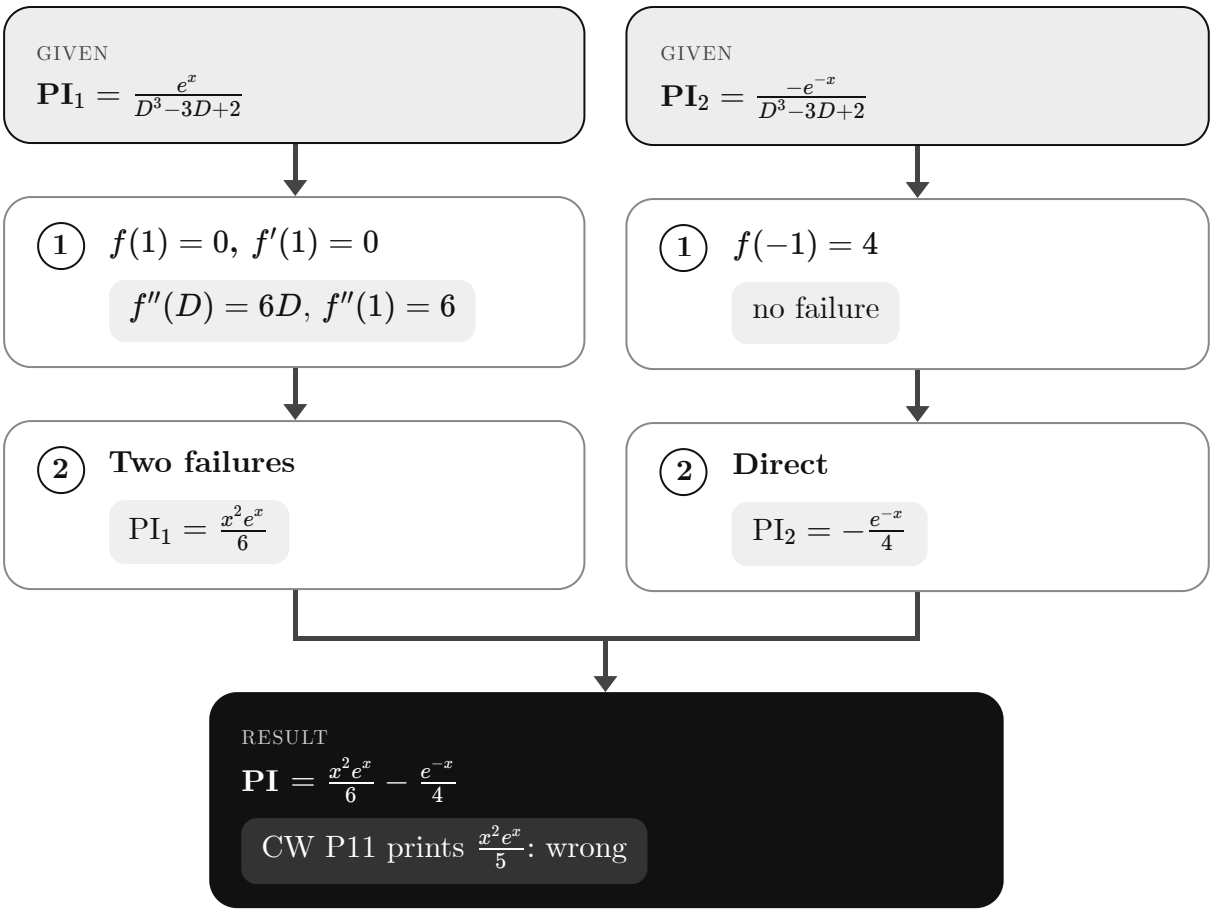
Flowcharts



Remark. If the replaced value is a root of f, climb the failure ladder (F3.1) for that term only.

Figure F4.1. What replaces D? ESA ★★★

S003·8 · S003·9 · S003·10 · T003 00:13:50 · T003 00:16:48



Remark. $2\sinh x = e^x - e^{-x}$. The minus sign belongs to PI_2 (slide 003·17 writes $\mathrm{PI}_2 = \frac{e^{-x}}{4}$ and subtracts later).

Figure F4.2. Two terms, two routes: $\frac{2\sinh x}{D^3-3D+2}$ (slides 003·16–17) ESA ★★

S003·16 · S003·17 · T004a 00:05:28 · CW U2 P11

Right-hand sides reducible to Type 1

X	Rewrite as	PI (no failure)
k	ke^{0x}	$\frac{k}{f(0)}$
$\sinh ax$	$\frac{1}{2}(e^{ax} - e^{-ax})$	$\frac{1}{2}\left[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}\right]$
$\cosh ax$	$\frac{1}{2}(e^{ax} + e^{-ax})$	$\frac{1}{2}\left[\frac{e^{ax}}{f(a)} + \frac{e^{-ax}}{f(-a)}\right]$
a^x	$e^{(\log a)x}$	$\frac{a^x}{f(\log a)}$
a^{-x}	$e^{-(\log a)x}$	$\frac{a^{-x}}{f(-\log a)}$

S003·9 · S003·10 · T003 00:13:50 · T003 00:16:48

The Type 1 term inside every ESA question

Paper, Q	$f(D)$	Type 1 term	Its PI	Full solution
Dec-23 3a (= May-24 3b, Jun-24 2a)	$(D-2)^2$	$8e^{2x}$	$4x^2e^{2x}$ (W3.5)	class 004
Jul-23 3b	D^2-5D+6	e^{4x}	$\frac{1}{2}e^{4x}$ (W3.6)	class 005
Jan-24 s1 2d	D^2+4	2^{-x}	$\frac{2^{-x}}{(\log 2)^2+4}$ (W4.5)	class 005
Jan-24 s2 2d	$(D-1)(D-2)(D-3)$	e^{2x}	$-xe^{2x}$ (W3.4)	class 005
Jan-24 s3 2c	$D(D-1)^2$	$\frac{1}{2}$ (from $\sin^2 x$)	$\frac{x}{2}$ (W4.7)	class 006
Jan-24 s4 2c	$(D-1)^2(D+1)^2$	$\frac{1}{2}$, 2^{-x}	$\frac{1}{2}+\frac{2^{-x}}{((\log 2)^2-1)^2}$ (W4.6)	class 005

ADDED FOR CLARITY

ESA Dec-23 3a · ESA May-24 3b · ESA Jun-24 2a · ESA Jul-23 3b · ESA Jan-24 s1 2d · ESA Jan-24 s2 2d · ESA Jan-24 s3 2c · ESA Jan-24 s4 2c

Important points and why

1. Rewrite the right-hand side as a sum of exponentials first, then take each term separately: $\mathrm{PI} = \mathrm{PI}_1 + \mathrm{PI}_2 + \ldots$ ESA ★★★ S003·12 · S003·16 · T003 00:19:31 · T004a 00:05:00
2. A constant is ke^{0x}: replace D by 0. If 0 is a root ($f(D)$ has no constant term), use the failure ladder with $a=0$: $\frac{1/2}{D^3-2D^2+D} = \frac{x}{2}$. ESA ★★ S003·9 · T003 00:14:25 · T004a 00:09:42 · T004a 00:10:14 · ESA Jan-24 s3 2c
3. For a^x replace D by $\log a$, for a^{-x} by $-\log a$. Leave $\log 2$ symbolic: $\frac{2^{-x}}{(\log 2)^2+4}$. ESA ★★ S003·10 · T003 00:16:34 · T003 00:16:48 · ESA Jan-24 s1 2d
4. $\sin^2\frac{x}{2} = \frac{1-\cos x}{2}$ and $\sin^2 x = \frac{1-\cos 2x}{2}$ hide a constant $\frac{1}{2}$: a Type 1 term inside a Type 2 question. ESA ★★ ESA Jan-24 s4 2c · ESA Jan-24 s3 2c · T004a 00:09:26
5. $\sinh$ and $\cosh$ terms may fail on one exponential only: in $\frac{2\sinh x}{D^3-3D+2}$ the e^x part fails twice, the e^{-x} part not at all. ESA ★★ S003·16 · S003·17 · T004a 00:05:28 · CW U2 P11

Why is the exponent of a^x equal to $\log a$?
$a^x = e^{\log(a^x)} = e^{x\log a}$, an exponential e^{bx} with $b = \log a$. Type 1 then replaces D by $b = \log a$. S003·10 · T003 00:16:11

Exam trap. a^x: replace D by $\log a$, not by a. $\frac{2^{-x}}{D^2+4} = \frac{2^{-x}}{(\log 2)^2+4}$, not $\frac{2^{-x}}{8}$. ADDED FOR CLARITY S003·10 · ESA Jan-24 s1 2d

Exam trap. CW P11's printed answer $\frac{x^2e^x}{5}$ is wrong: $f''(D) = 6D$, $f''(1) = 6$, so $\frac{x^2e^x}{6}$ (slide 003·17 and T004a agree). CW U2 P11 · S003·17 · T004a 00:05:28

Exam trap. Slide 003·17 calls $\text{PI}_2 = \frac{e^{-x}}{4}$ but then subtracts it; keep the sign inside: $\text{PI}_2 = \frac{-e^{-x}}{f(-1)} = -\frac{e^{-x}}{4}$. S003·16 · S003·17

Exam trap. Expand before splitting: $(e^{-x}+2)^2 = e^{-2x} + 4e^{-x} + 4$, and $e^{-x}\cosh x = \frac{1+e^{-2x}}{2}$ adds to both the constant and the e^{-2x} term (CW P12). ADDED FOR CLARITY CW U2 P12

Worked examples

W4.1 · Constant: $D^2y + 2Dy + 5y = 1$ (slide 003·14) ESA ★★

GIVEN Find the PI of $(D^2 + 2D + 5)y = 1$.

STEP 1 $1 = e^{0x}$, so $a = 0$.

STEP 2 $f(0) = 0 + 0 + 5 = 5 \neq 0$.

ANSWER $\boxed{\text{PI} = \frac{1}{5};\ y = e^{-x}(c_1\cos 2x + c_2 \sin 2x) + \frac{1}{5}}$

CHECK sympy dsolve ✓ (AE roots $-1 \pm 2i$).

S003·14 · T004a 00:02:06 · T004a 00:02:23

W4.2 · $(D^2 - 2D + 2)y = a^{-x} + e^x$ (slides 003·11–12) ESA ★★★

GIVEN Solve $(D^2 - 2D + 2)y = a^{-x} + e^x$.

STEP 1 AE $m^2 - 2m + 2 = 0$, $m = 1 \pm i$: $y_c = e^x(c_1\cos x + c_2 \sin x)$.

STEP 2 $\frac{a^{-x}}{D^2-2D+2}$: replace D by $-\log a$: denominator $(-\log a)^2 - 2(-\log a) + 2 = (\log a)^2 + 2\log a + 2$.

STEP 3 $\frac{e^x}{D^2-2D+2}$: replace D by 1: $1 - 2 + 2 = 1$, so e^x.

STEP 4 $y = y_c + y_p$.

ANSWER $\boxed{y = e^x(c_1\cos x + c_2 \sin x) + \frac{a^{-x}}{(\log a)^2 + 2\log a + 2} + e^x}$

CHECK Substituting y_p back (sympy, symbolic $a > 0$) ✓; dsolve with $a = 2$ ✓. $(\log a)^2 + 2\log a + 2 = (\log a + 1)^2 + 1 > 0$, so no failure for any a. The slide stops before simplifying.

S003·11 · S003·12 · T003 00:17:12 · T003 00:18:30 · T003 00:20:01 · T003 00:20:18

W4.3 · $2\sinh x$: $y''' - 3y' + 2y = 2\sinh x$ (slides 003·16–17, CW P11) ESA ★★★

GIVEN Solve $(D^3 - 3D + 2)y = 2\sinh x$.

STEP 1 AE $m^3 - 3m + 2 = (m-1)^2(m+2) = 0$: $y_c = (c_1 + c_2 x)e^x + c_3 e^{-2x}$.

STEP 2 $2\sinh x = e^x - e^{-x}$, so PI $= \frac{e^x}{D^3 - 3D + 2} - \frac{e^{-x}}{D^3 - 3D + 2}$.

STEP 3 e^x: $f(1) = 0$; $f'(D) = 3D^2 - 3$, $f'(1) = 0$; $f''(D) = 6D$, $f''(1) = 6$: $\frac{x^2 e^x}{6}$.

STEP 4 e^{-x}: $f(-1) = -1 + 3 + 2 = 4$: $\frac{e^{-x}}{4}$, subtracted.

ANSWER $\boxed{y = (c_1 + c_2 x)e^x + c_3 e^{-2x} + \frac{x^2 e^x}{6} - \frac{e^{-x}}{4}}$

CHECK sympy dsolve and checkodesol ✓. The CW sheet prints $\frac{x^2 e^x}{5}$, which leaves $\frac{e^x}{5}$ over: CW answer wrong. Slide 003·17 labels $\mathrm{PI}_2 = \frac{e^{-x}}{4}$ but subtracts it; the final line is right.

S003·16 · S003·17 · T004a 00:04:43 · T004a 00:05:28 · T004a 00:05:47 · CW U2 P11

W4.4 · Homework CW P12: $(e^{-x} + 2)^2 + e^{-x}\cosh x$ ESA ★★

GIVEN Solve $(D^4 + D^3 - 3D^2 - 5D - 2)y = (e^{-x} + 2)^2 + e^{-x}\cosh x$.

STEP 1 AE: $m = -1$ is a triple root and $m = 2$: $f(D) = (D+1)^3(D-2)$, $y_c = (c_1 + c_2 x + c_3 x^2)e^{-x} + c_4 e^{2x}$.

STEP 2 RHS $= e^{-2x} + 4e^{-x} + 4 + \frac{1 + e^{-2x}}{2} = \frac{3}{2}e^{-2x} + 4e^{-x} + \frac{9}{2}$.

STEP 3 $\frac{3}{2}e^{-2x}$: $f(-2) = 16 - 8 - 12 + 10 - 2 = 4$: $\frac{3}{8}e^{-2x}$.

STEP 4 $4e^{-x}$: $f(-1) = f'(-1) = f''(-1) = 0$; $f'''(D) = 24D + 6$, $f'''(-1) = -18$: $\frac{4x^3 e^{-x}}{-18} = -\frac{2}{9}x^3 e^{-x}$.

STEP 5 $\frac{9}{2}$: $f(0) = -2$: $-\frac{9}{4}$.

ANSWER $\boxed{y = (c_1 + c_2 x + c_3 x^2)e^{-x} + c_4 e^{2x} + \frac{3}{8}e^{-2x} - \frac{2}{9}x^3 e^{-x} - \frac{9}{4}}$

CHECK sympy dsolve and checkodesol ✓; the CW answer agrees. Shortcut E3.5: $\frac{4x^3 e^{-x}}{3!\,\varphi(-1)}$ with $\varphi(-1) = -3$.

CW U2 P12

W4.5 · 2^{-x} part of ESA Jan-24 s1 2d

ESA ★★★

GIVEN Find the PI of 2^{-x} in $(D^2+4)y = x^2 + \cos 2x + 2^{-x}$.

STEP 1 $2^{-x} = e^{-(\log 2)x}$: replace D by $-\log 2$.

STEP 2 $f(-\log 2) = (\log 2)^2 + 4 \neq 0$.

ANSWER $\boxed{\text{PI}_3 = \frac{2^{-x}}{(\log 2)^2+4} \approx 0.2232 \cdot 2^{-x}}$

CHECK Substituting back (sympy) ✓. The x^2 (Type 3) and $\cos 2x$ (Type 2 failure) parts and the full answer are in the class 005 pack.

ESA Jan-24 s1 2d

W4.6 · 2^{-x} and the constant, ESA Jan-24 s4 2c (CW P22)

ESA ★★★

GIVEN Find the Type 1 parts of the PI of $(D-1)^2(D+1)^2 y = \sin^2\frac{x}{2} + 2^{-x} + x$.

STEP 1 $\sin^2\frac{x}{2} = \frac{1}{2} - \frac{1}{2}\cos x$: the constant $\frac{1}{2}$ is Type 1 with $a=0$; $f(0) = (1)(1) = 1$: $\frac{1}{2}$.

STEP 2 2^{-x}: $f(-\log 2) = (-\log 2 - 1)^2(-\log 2 + 1)^2 = \left((\log 2)^2 - 1\right)^2 = (\log 2)^4 - 2(\log 2)^2 + 1$.

STEP 3 So $\frac{2^{-x}}{(\log 2)^4 - 2(\log 2)^2 + 1}$ ($\approx 3.705 \cdot 2^{-x}$).

ANSWER $\boxed{\frac{1}{2} + \frac{2^{-x}}{(\log 2)^4 - 2(\log 2)^2 + 1}}$

CHECK Both substitute back (sympy) ✓, and match CW P22's printed terms. The $\cos x$ and x parts and the full answer are in the class 005 pack.

ESA Jan-24 s4 2c · CW U2 P22

W4.7 · Constant with a zero root, ESA Jan-24 s3 2c

ESA ★★

GIVEN Find the PI of the constant in $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos 2x$ for $(D^3 - 2D^2 + D)y = x^2 e^{-x} + \sin^2 x$.

STEP 1 $\frac{1/2}{D^3 - 2D^2 + D}$ with $a=0$: $f(0) = 0$, failure (0 is a root).

STEP 2 $f'(D) = 3D^2 - 4D + 1$, $f'(0) = 1$: $\frac{1}{2} \cdot \frac{x}{1}$.

ANSWER $\boxed{\text{PI} = \frac{x}{2}}$

CHECK $(D^3 - 2D^2 + D)\frac{x}{2} = \frac{1}{2}$ ✓. The $x^2 e^{-x}$ (Type 4) and $\cos 2x$ parts are in the class 006 pack.

ESA Jan-24 s3 2c · T004a 00:09:42 · T004a 00:10:01 · T004a 00:10:14

W4.8 · $\cosh 2x$ and 3^x ESA ★

GIVEN Find $\frac{\cosh 2x}{D^2-1}$ and $\frac{3^x}{D-1}$.

STEP 1 $\frac{\cosh 2x}{D^2-1} = \frac{1}{2}\left[\frac{e^{2x}}{4-1} + \frac{e^{-2x}}{4-1}\right] = \frac{\cosh 2x}{3}$.

STEP 2 $\frac{3^x}{D-1}$: replace D by $\log 3$: $\frac{3^x}{\log 3-1}$.

ANSWER $\frac{1}{3}\cosh 2x$ and $\frac{3^x}{\log 3-1}$

CHECK Both substitute back (sympy) ✓.

ADDED FOR CLARITY S003·10

Past-paper questions

ESA Jan 2024 · 5 marks · UE23MA141A set 1, Q2.d · solved in W4.5

Solve $(D^2+4)y = x^2 + \cos 2x + 2^{-x}$. Type 1 part: $\frac{2^{-x}}{(\log 2)^2+4}$. Full solution: class 005 pack.

ESA Jan 2024 · 6 marks · UE23MA141A set 4, Q2.c (= CW P22) · solved in W4.6

Solve $[(D-1)^2(D+1)^2]y = \sin^2\frac{x}{2} + 2^{-x} + x$. Type 1 parts: $\frac{1}{2}$ and $\frac{2^{-x}}{(\log 2)^4-2(\log 2)^2+1}$. Full solution: class 005 pack.

ESA Jan 2024 · 6 marks · UE23MA141A set 3, Q2.c (= CW P18) · solved in W4.7

Solve $(D^3-2D^2+D)y = x^2e^{-x} + \sin^2 x$. Type 1 part: the constant $\frac{1}{2}$ gives $\frac{x}{2}$ (failure at $a=0$). Full solution: class 006 pack.

SHORT-NOTE RECAP

Constants: $\frac{k}{f(D)} = \frac{k}{f(0)}$ (if $f(0)=0$: $\frac{xk}{f'(0)}$). $\frac{\sinh ax}{f(D)} = \frac{1}{2}[\frac{e^{ax}}{f(a)} - \frac{e^{-ax}}{f(-a)}]$, cosh with +. $\frac{a^x}{f(D)} = \frac{a^x}{f(\log a)}$, $\frac{a^{-x}}{f(D)} = \frac{a^{-x}}{f(-\log a)}$. CW P11: $\frac{x^2e^x}{6} - \frac{e^{-x}}{4}$ (the sheet's $\frac{x^2e^x}{5}$ is wrong); CW P12: $\frac{3}{8}e^{-2x} - \frac{2}{9}x^3e^{-x} - \frac{9}{4}$.

A Symbol glossary

Symbol	Name	Meaning	Appears in
D	Differential operator	$\frac{d}{dx}$	all
$f(D)$	Linear differential operator	Polynomial in D with constant coefficients	E1.1–E4.4
$\frac{1}{f(D)}$	Inverse differential operator	Gives the PI: $f(D)\{\frac{1}{f(D)}X\} = X$	E1.1, E1.2
X	Right-hand side	Function of x (capital X, not the variable x)	E1.1–E2.3
y_c, CF	Complementary function	General solution of $f(D)y = 0$	E1.2
y_p, PI	Particular integral	$\frac{1}{f(D)}X$, no arbitrary constants	E1.2
a	Exponent (Type 1)	In e^{ax}; replaces D. Clash: the base in a^x (exponent $\log a$)	E3.1–E3.4, E4.4
$f'(a),\ f''(a)$	Derivatives of f	$\frac{d}{dD}f(D)$, $\frac{d^2}{dD^2}f(D)$ at $D = a$	E3.2–E3.4
k, K	Constant RHS	Also the multiplicity k in E3.5 (added for clarity)	E4.1, E3.5
$\varphi(D)$	Cofactor	$f(D) = (D-a)^k\varphi(D)$, $\varphi(a) \neq 0$ (added for clarity)	E3.5
$\log a$	Natural logarithm	Exponent of $a^x = e^{x\log a}$	E4.4
$\sinh,\ \cosh$	Hyperbolic functions	$\frac{e^{ax} \mp e^{-ax}}{2}$	E4.2, E4.3

B Sources, coverage and gaps

Source	Path	Use
Deck 003 (.pptx rendered; .slides.md; .slides.txt)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/Slides/003_Particular_integral_of_Type_1__f4ba122af8cfae89.*	18 slides; content on 3–17 (every equation read from the rendered images)
Transcript 003 (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.*	21 min; slides 3–12
Transcript 004a, Type 1 part (all 4 formats)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/004_Particular_integral_Type_1_and_Type_2__54bec5745a32a447.*	00:00:00–00:06:04: Type 1 recap and slides 14–17 worked; 00:09:26–00:10:35: constant term of $\sin^2 x$ (failure at $a = 0$)
Unit 1 & 2 class-work sheet (UE24MA141A)	pesuc/First_Year_8enterprise/Files/Math 1/CW/Math_Unit1&2_CW__1jKDN6W8iw.pdf	Unit 2 problems 10–12 (page 5 rendered at 250 dpi to read the answers); Type 1 terms of problems 21 and 22
ESA papers, UE22MA141A and UE23MA141A (8 papers)	pesuc/First_Year_8enterprise/Files/Math 1/Papers/ESA Papers/UE2[23]MA141A/*.pdf	Type 1 parts of 6 distinct questions (table C4.2), via the shared ESA map
Videos 003 and 004 (.mp4)	pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/003_Particular_integral_of_standard_functions_-_Type_1__22d81006814d93bf.mp4	Not opened: every slide equation is readable and the transcripts raise no question the slides cannot answer

Eight past ESA papers were checked: UE22MA141A Dec 2023, Jul 2023 and May 2024, and UE23MA141A Jan 2024 sets 1–4 and June 2024 (19 distinct Unit 2 questions, 24 occurrences). No question is Type 1 alone, but a Type 1 term sits inside 6 distinct questions (8 occurrences): $8e^{2x}$ with a double failure in Dec-23 3a (= May-24 3b = Jun-24 2a), e^{4x} in Jul-23 3b, 2^{-x} in Jan-24 set 1 2d and set 4 2c, e^{2x} with a single failure in set 2 2d, and the constant $\frac{1}{2}$ of $\sin^2$ in set 3 2c (a failure at $a = 0$) and set 4 2c. Stars: the rule and its first two failure cases ★★★; a^x, constants, $\sinh$ / $\cosh$ and $\frac{1}{D-a}$ ★★ (asked once or twice, or class-work); the third failure (x^3) ★ (lecture only, never asked).

Known gaps, conflicts and corrections

1. CW P11 answer is wrong: it prints $\frac{x^2e^x}{5}$; the correct term is $\frac{x^2e^x}{6}$, since $f''(D) = 6D$ gives $f''(1) = 6$ (slide 003·17 and T004a 00:05:28 agree with 6). Substituting the CW answer back leaves $\frac{e^x}{5}$ over (sympy).
2. CW P10 answer is misprinted: it reads $(c_1e^{-x} + c_2e^{4x}x) - \frac{e^x}{6} + \frac{4xe^x}{5}$. The correct answer is $c_1e^{-x} + c_2e^{4x} - \frac{e^x}{6} + \frac{4}{5}xe^{4x}$: the x belongs to the PI, and its exponential is e^{4x}.

3. CW P12 (homework) answer is right: $(c_1 + c_2x + c_3x^2)e^{-x} + c_4e^{2x} + \frac{3}{8}e^{-2x} - \frac{2}{9}x^3e^{-x} - \frac{9}{4}$ (sympy dsolve and checkodesol).
4. Slide 003·17 writes "In PI_2 ... $f(a) = 4$ then $PI_2 = \frac{e^{-x}}{4}$" but slide 003·16 defined $PI_2 = \frac{-1}{D^3-3D+2}e^{-x}$, which is $-\frac{e^{-x}}{4}$. The final line $\frac{x^2e^x}{6} - \frac{e^{-x}}{4}$ is right; only the sign bookkeeping slips. Slide 003·16 also writes "$f(D) = (D^3 - 3D + 2)y = e^x - e^{-x}$", mixing the operator and the equation.
5. Slide 003·12 stops at $y = y_c + y_p$ with y_p unsimplified; this pack finishes it: $\frac{a^{-x}}{(\log a)^2+2\log a+2} + e^x$ (the second denominator is $1 - 2 + 2 = 1$).
6. Slide 003·9 gives $\frac{K}{f(D)} = \frac{K}{f(0)}$ without the proviso $f(0) \neq 0$; the lecture adds it and the failure $\frac{xK}{f'(0)}$ (T003 00:14:25). Slide 003·13 says "similarly, it proceeds on when $f''(a) = 0$"; the lecture gives the next rung $\frac{x^3e^{ax}}{f'''(a)}$ (T003 00:13:19).
7. Slide spellings "arbitarary" (003·4) and "Complimentary function" (003·6) mean arbitrary and complementary. The lecture says $\frac{1}{D^2}X$ is "2 times integration" (T003 00:07:03): it means integrating twice, $\iint X\,dx\,dx$, not $2\int X\,dx$.
8. The rule for a root of multiplicity k in the form $\frac{x^ke^{ax}}{k!\,\varphi(a)}$ (with $f(D) = (D - a)^k\varphi(D)$) is added for clarity; it is equivalent to the slides' ladder.
9. No slide answer in this class is wrong: slides 003·12, 14, 15 and 16–17 were recomputed (sympy, substitution and dsolve).
10. ESA: no question is Type 1 alone; the full mixed questions are solved in the class 004 (Dec-23 3a), 005 (Jul-23 3b, Jan-24 s1 2d, s2 2d, s4 2c) and 006 (Jan-24 s3 2c) packs. Here only their Type 1 terms are worked.
11. Transcript recognition errors were corrected silently: "f of b"/"f of df y"/"1 by eight f of d" → $f(D)y$, $\frac{1}{f(D)}$; "E par ax"/"e po"/"e bar" → e^{ax}; "human differential equation" → given differential equation; "R. constant" → arbitrary constant; "important resources" → important results; "sine hyperbolic eggs" → $\sinh ax$; "psi beta x" → $\sin \beta x$; "plus r minus" → $\pm$; "login" → $\log a$; "d minus e"/"d minus p" → $D - a$, $D - b$; in T004a "k power X" → e^{ax}, "h square" → x^2, "replace d by gate" → replace D by 0.

Coverage

Source	Citations	Slides cited (slide × citations)
S003	92	4×6, 5×2, 6×7, 7×11, 8×14, 9×7, 10×11, 11×2, 12×5, 13×10, 14×2, 15×5, 16×5, 17×5
CW	13	
ESA	37	
T003	83	
T004a	27	

Slides 1, 2, 18: title and closing; slide 3: class content. The lecture covers slides 4–12; slides 13–17 are worked in the class 004 problem lecture (T004a). Bucket numbers are not contiguous (decks say SESSION 1–11; 007, 008, 010, 012, 013 do not exist; 009 = session 7, 011 = 8 and 9, 014 = 10, 015 = 11); this deck is SESSION 3.